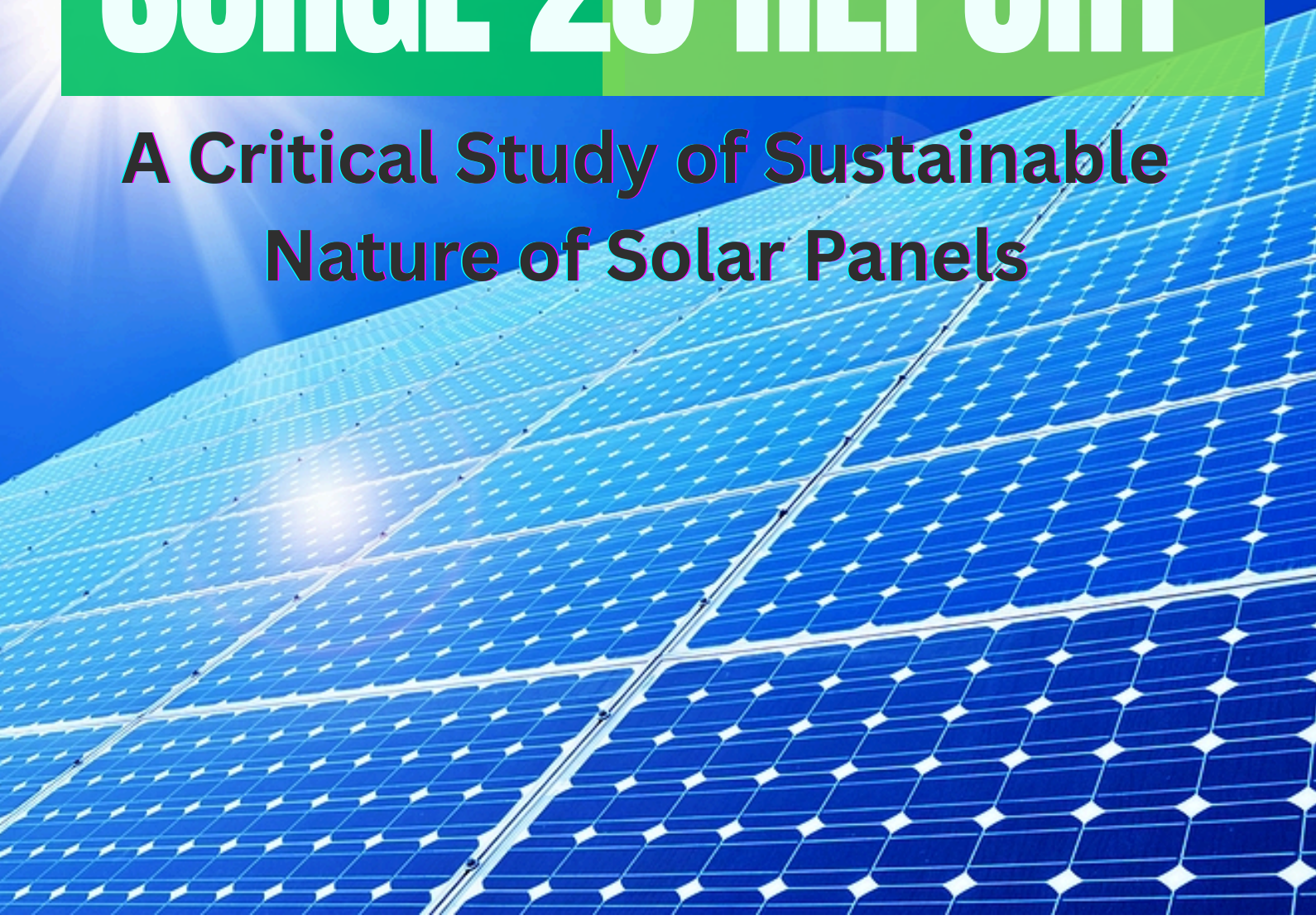
# SURGE'25 REPORT

## A Critical Study of Sustainable Nature of Solar Panels



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# PROBLEM STATEMENT

While solar photovoltaic (PV) technology is widely recognized as a green and renewable source of energy, but its complete environmental footprint across the panel's life cycle remains a growing concern. The production, deployment, and end-of-life management of solar panels involve energy-intensive processes, use of rare and sometimes toxic materials, and currently inadequate recycling methods. Even large scale installations of solar panels cause climate change globally. As global solar installations continue to rise rapidly, these sustainability challenges-often overlooked-pose significant environmental risks in the long term. Therefore, there is a critical need to holistically evaluate the complete sustainability of solar panels, considering not only their low operational emissions but also the hidden impacts from manufacturing to disposal. This study addresses the gap by assessing the overall sustainability of solar panels and identifying strategies to improve their environmental and economic viability.

# ABSTRACT

Over a decade, we saw a drastic change in climate due to increased usage of fossil fuels as source of energy. So as an alternative we are turning towards renewable and green sources. Here in the Solar panels came into the picture. Solar energy is energy derived from the sun's radiation. It is widely celebrated as a renewable and green source. Therefore , in recent years energy production through Solar Panels is increasing rapidly across the globe. This accounts to its emission free operations and cheap than other traditional sources. Solar energy is much better than fossil fuels in terms of environmental impact and long-term sustainability. Even when compared to wind and nuclear energy, it is more beneficial in terms of safety, decentralization, and ecological harmony.

However, alongside its many advantages, there are also several overlooked criticisms of solar energy. It is heavily dependent on industrial manufacturing, mining of rare materials, and large-scale land occupation — all of which leave a significant environmental footprint. According to model studies, large-scale solar farms can even alter global climatic patterns and cause the solar heat island effect. Additionally, solar systems face issues such as intermittency, storage dependency, and the growing concern of e-waste from disposed panels. These issues raise critical questions about the true sustainability of solar panels when examined beyond its surface-level benefits.

Hence , energy production through solar panels is not completely renewable and its negative effects may impact environment significantly if those issues are overlooked since solar panel installation is increasing steadily. Therefore necessary precautions such as green production techniques, proper management of toxic waste, combining solar panels with farming and biodiversity-sensitive cite planning can taken immediately.

**Key Words:** Green Energy, Solar Energy , Carbon footprint, Global Warming, Climate Change , Photovoltaic cell, Sustainability, Solar Heat Island Effect

# INTRODUCTION

Day by day, the growing population is escalating energy demands globally. With rising global energy demands and growing concerns about climate change due to burning of fossil fuels, solar energy has emerged as a leading green energy source. However, despite its renewable nature, emission free operation and cheap price, the complete sustainability of solar panels depends on their complete life cycle — from raw material extraction to disposal. While solar panels appear to be a remarkable source of clean energy, they come with hidden limitations that are frequently overlooked and cause significant environmental impact.

This study aims to assess the overall sustainability of solar panels by addressing key questions such as:

- Are solar panels truly green source of energy?
- How solar panels cause damage to the nature?
- What are the hidden environmental concerns raised by large-scale installation of solar panels?
- How can sustainability of solar panels be improved?



# LITERARY REVIEW

Solar photovoltaic (PV) tchnology has been extensively researched as a greener alternative to fossil fuels, owing to its emission free operation and decrease reliance on non-renewable sources. Numerous studies have performed life cycle assessments (LCA) to evaluate the true environmental impact of solar panels beyond their operational phase. These assessments consider emissions and energy usage during manufacturing, usage, and disposal phases.
Fthenakis and Kim (2009) conducted one of the earliest comprehensive LCAs of solar PV systems, estimating life cycle emissions of monocrystalline silicon panels at 40–60 $gCO_2$/kWh, a stark contrast to coal (800–1000 $gCO_2$/kWh) and natural gas (~500 $gCO_2$/kWh). IEA PVPS (2020) reports show that with improvements in panel efficiency and manufacturing technologies, the energy payback time (EPBT) for panels has dropped to 1–2 years, while lifetimes typically extend beyond 25 years.
However, recent literature increasingly focuses on the underreported **drawbacks associated with solar panels:**

**a) Emissions During Manufacturing**

Studies such as those by IRENA (2022) and Held et al. (2021) reveal that manufacturing processes — especially for silicon purification, wafer production, and module assembly — are energy-intensive and often rely on coal-based electricity, particularly in countries like China. These upstream emissions can account for more than 75% of a panel's total life cycle emissions.

**b) Toxic Materials and Waste**

Solar panels contain hazardous materials such as cadmium, lead, and antimony, which can pollute soil and water if not properly managed (Cucchiella et al., 2015; Kang et al., 2014). Additionally, the purification of silicon for solar cells produces silicon tetrachloride ($SiCl_4$) — a highly reactive and toxic byproduct. Improper disposal of $SiCl_4$ has caused severe environmental damage, especially in poorly regulated regions (Lu et al., 2015; Fthenakis, 2009). These concerns affect both thin-film (e.g., CdTe) and crystalline silicon technologies, underlining the need for stricter recycling and waste-handling protocols.

**c) End-of-Life Management Challenges**

The issue of panel waste is a growing concern. A report by the International Renewable Energy Agency (IRENA, 2016) warns that by 2050, solar panel waste could reach 78 million tonnes globally. Current recycling technologies are costly and inefficient, recovering only 60–70% of the material. Studies urge for the development of closed-loop recycling systems and stronger extended producer responsibility (EPR) frameworks.

**d) Solar Heat Island Effect**

Barron-Gafford et al. (2016) present empirical evidence showing that large-scale solar installations can increase local temperatures, a phenomenon termed the Photovoltaic Heat Island Effect (PVHI). Similar to the urban heat island effect, PVHI results from changes in surface albedo, air circulation, and heat retention caused by expansive solar arrays. This localized heating can alter temperature, humidity, and plant productivity, particularly in desert and semi-arid regions, potentially affecting local ecosystems and microclimates.

**e) Ecological and Biodiversity Impacts**

Recent studies, including those by Hernandez et al. (2014), emphasize the land-use conflicts arising from utility-scale solar farms. Solar installations can fragment habitats, displace wildlife, and degrade ecosystems, whether built on forests, grasslands, migratory corridors, or even deserts—where they disturb fragile ecosystems and specialized species. Solar energy's ecological footprint is often underestimated in national energy policies.

**f) Global Climate Implications**

Research published in Scientific Reports (Li et al., 2016) suggests that large-scale deployment of solar panels could potentially alter regional climate patterns, cloud cover, and wind circulation, especially in concentrated areas. While this effect is still under investigation, it raises concerns about the broader climatic feedback of renewable infrastructures.

# METHODOLOGY

This study adopts a qualitative research methodology based on a comprehensive review of published research articles, life cycle assessment (LCA) reports, and global energy databases. The analysis follows a production to disposal LCA framework, evaluating the environmental impacts of solar photovoltaic (PV) panels across four key life stages:

- **Material extraction and manufacturing**
- **Panel transportation and installation**
- **Operational phase (electricity generation)**
- **End-of-life disposal and recycling**

Initially, the study focuses on the positive sustainability indicators of solar PV systems. This includes benchmarking their performance against conventional fossil fuel sources (e.g., coal, natural gas), with attention to:

- **Carbon emissions ($gCO_2eq/kWh$)**
- **Energy payback time (EPBT)**
- **Lifetime energy output**

The study also examines critical sustainability challenges that are less frequently emphasized. These include:

- **High emissions and energy use during manufacturing**
- **Toxic byproducts and rare material extraction**
- **Inadequate recycling infrastructure and future e-waste concerns**
- **Solar Heat Island Effect (SHIE) and land-use impacts**
- **Potential disruptions to local ecosystems, biodiversity, and regional climate**

All data and insights are drawn from a synthesis of over 20 scholarly articles, institutional reports (e.g., IEA, IRENA), and existing LCA databases. No primary experimental work is conducted; rather, the study relies on secondary data to evaluate the environmental and sustainability profile of solar PV systems holistically

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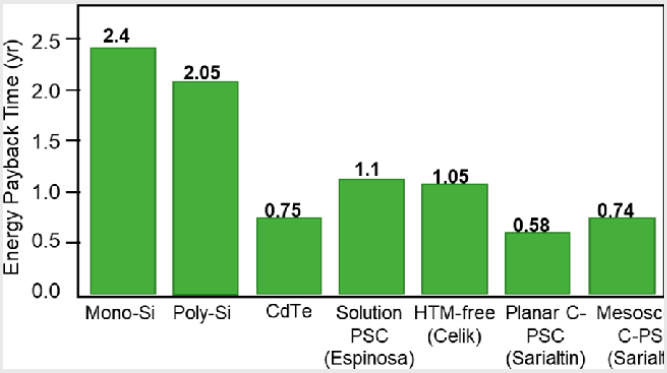
# DISCUSSION

## Positive Sustainability Metrics

The literature constantly shows that solar PV panels emit significantly lower $CO_2$ over their lifetime compared to fossil fuels. Average life cycle emissions for monocrystalline silicon panels range between 40–60g $CO_2$ eq/kWh, whereas coal-based electricity exceeds 800–1000g $CO_2$ eq/kWh.

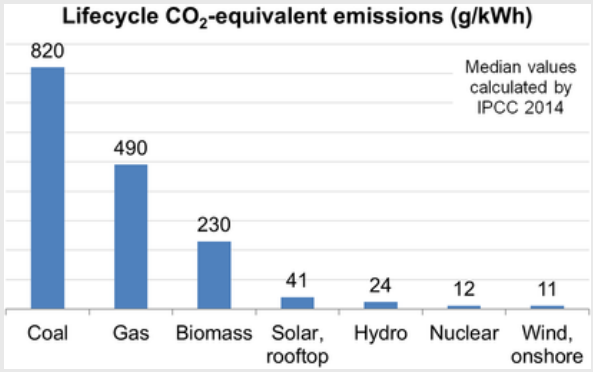
Moreover, energy payback times have improved drastically with modern PV technology — dropping to 1–2 years, while panel lifetimes often exceed 25 years.

Such figures confirm that from a use-phase outlook, solar panels are a highly sustainable alternative to fossil fuels.




Energy payback time (EPBT) comparison of different C-PSC (HTM-free) architectures with the literature and first and second generation PV technologies. Planar PSCs with the carbon electrode show the lowest EPBT (0.58 yr) compared to other technologies. In particular, C-PSCs exhibit approximately 4 times lower EPBT than mono-and poly-silicon solar cells. Reproduced with permission from ref 73. Copyright 2020 AIP.

**Source:** Energy payback time (EPBT) comparison of different C-PSC HTM-free architectures…" from ResearchGate, originally published in [Author(s)] (2023). Journal Name




Carbon emissions of different energy sources in the whole lifetime

**Source:** Solar.com (2024)

## Sustainability Drawbacks

However, deeper analysis reveals several critical environmental concerns that are often neglected:

### a) Emissions during Manufacturing

Production of silicon wafers involves high-temperature energy intensive processes using fossil-fuel energy.
Studies estimate that up to 80% of total life cycle emissions occur before the panel is even installed.

### b) Toxic Waste and Rare Materials

Panels often contain lead, cadmium, antimony, and silver, which are toxic and pose risks to soil and groundwater if not properly disposed.
Mining of rare materials (e.g., quartz, indium, tellurium) causes local ecological degradation. Additionally, in the purification of silicon , toxic $SiCl_4$ is produced , if not managed properly can pollute water and soil.



### c) End-of-Life  Waste

Globally, an estimated 78 million tonnes of solar panel waste could be accumulated.
Current recycling systems are underdeveloped, recovering only low-value materials like aluminum and glass, and often incinerating or landfilling the rest.

### d) Solar Heat Island Effect

Large-scale solar farms may increase local ambient temperatures by 2–5°C, altering local microclimates — similar to the Urban Heat Island effect.
This thermal alteration can cause harm to local flora and insect biodiversity

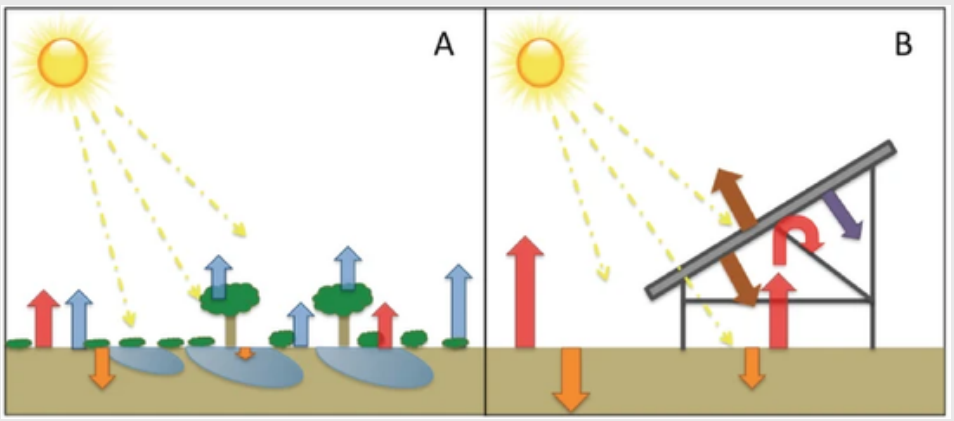


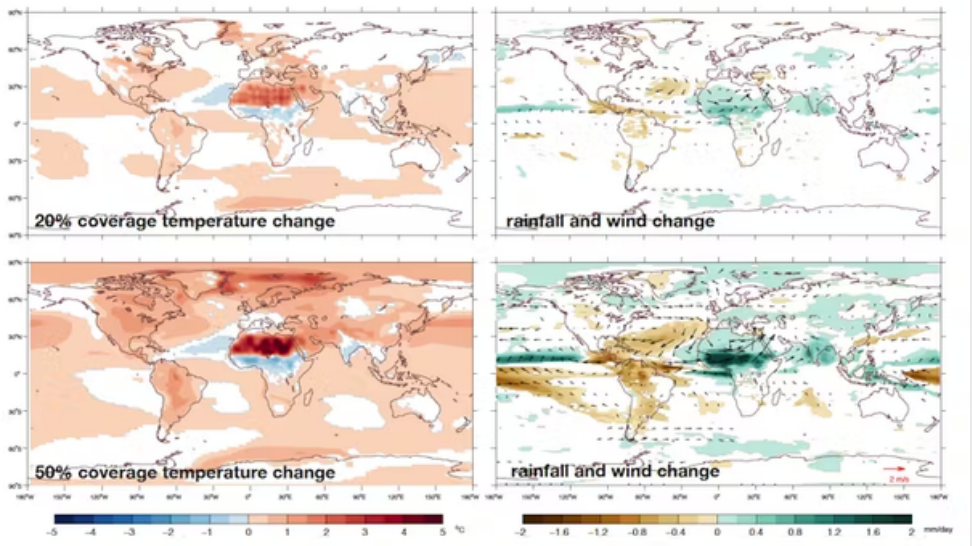

Illustration of midday energy exchange.
Assuming equal rates of incoming energy from the sun, a transition from (A) a vegetated ecosystem to (B) a photovoltaic (PV) power plant installation will significantly alter the energy flux dynamics of the area. Within natural ecosystems, vegetation reduces heat capture and storage in soils (orange arrows), and infiltrated water and vegetation release heat-dissipating latent energy fluxes in the transition of water-to-water vapor to the atmosphere through evapotranspiration (blue arrows). These latent heat fluxes are dramatically reduced in typical PV installations, leading to greater sensible heat fluxes (red arrows). Energy re-radiation from PV panels (brown arrow) and energy transferred to electricity (purple arrow) are also shown.

## e) Impact on Biodiversity and Land Use

Utility-scale solar farms require vast land, often leading to:
Habitat fragmentation
Displacement of local species
Disruption of migratory pathways (e.g., for birds or mammals)
In arid or sensitive regions, large-scale installations may even disrupt water cycles and cause desertification.

## f) Global Climate Alterations

A few simulation-based studies (e.g., Millstein & Menon, Nature Sci. Rep. 2016) suggest that large-scale deployments could potentially alter wind patterns, cloud formation, and surface albedo, impacting regional climate systems.




Global temperature, rainfall and surface wind changes in simulations with 20 percent and 50 percent solar panel coverage of Sahara. Lu et al. (2021).// Image courtesy of The Conversation

## Critical Reflection

While solar panels are widely known for their low operational emissions and renewable nature, the narrow focus on use-phase sustainability risks creating blind spots in policy and public perception. The results of this study suggest the importance of a full life cycle outlook, integrating:

- Green production methods
- Circular process of production
- End-of-life management
- Biodiversity-sensitive site planning

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